Mandeep Singh

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November 06, 2026

Hiring Team
RBC
Toronto, ON

Re: 2027 Capital Markets, Global Equities, AI Engineer (16 Months, Co-op)

Dear Hiring Team,

I am writing to apply for the 2027 Capital Markets, Global Equities, AI Engineer Co-op with RBC's Algo Research team. Applying machine learning to trade execution, where models are judged by measurable outcomes rather than demos, is the kind of disciplined, data-driven engineering I want to build my career on, and a 16-month term from May 2027 to August 2028 would let me contribute to real research.

At InferenceSaver, I co-founded a production AI gateway and multimodal generation platform that reached 2.7K active site users and 43K tracked events in 2026. I built the Python and TypeScript services, databases, and integrations behind it, and I learned to treat correctness as non-negotiable: I traced a payment idempotency defect affecting 8 of 14 checkouts by reconciling transaction records against application state, then implemented retry-safe writes.

I am comfortable working with large datasets and Unix environments. I built Generic Alternatives on PostgreSQL, Docker, and background job queues for structured data processing, and I contribute to AgentMemory, an open-source TypeScript project with 29.7K+ GitHub stars spanning session recording and semantic retrieval. Coursework in operating systems and computer networks gives me the systems foundations the role lists, and I am actively building depth in PyTorch, scikit-learn, pandas, and the statistics behind feature and loss-function design.

Running two products has also taught me to explain technical tradeoffs to non-technical partners clearly. I would welcome the opportunity to work alongside RBC's traders and researchers on AI-driven execution. Thank you for considering my application.

Sincerely,

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Summary

Third-year Computing Science co-op student with strong Python, databases, Unix, and networking fundamentals, production AI integrations, and data-heavy backend systems. Seeking RBC Capital Markets' 16-month Global Equities AI Engineer Co-op (May 2027 to August 2028) with the Algo Research team.

Education

Thompson Rivers University Kamloops, BC

Bachelor of Computing Science (Co-op Program) *2023 – Expected April 2028*

- Third-year student focused on software engineering, information systems, and computer networks
- Relevant coursework: Computer Networks, Operating Systems, Web-Based Information Systems

Experience

Co-Founder & Software Engineer July 2025 – Present

InferenceSaver *Next.js, React, TypeScript, Convex, Stripe*

- Co-founded and developed a production **AI API gateway and multimodal generation platform** providing unified access to language, image, video, music, and voice models for **129 registered users**
- Built product interfaces, backend workflows, authentication, payments, affiliate systems, and model integrations supporting **$5.6K+ in confirmed platform credit allocations across 60 top-ups**
- Diagnosed a Stripe idempotency defect affecting **8 of 14 completed checkouts**, implemented retry-safe payment writes, and restored music/voice integrations used for **88 production generation jobs**

Co-Founder & Technical Lead March 2026 – Present

BecomeAfish *Next.js, TypeScript, PostgreSQL, Stripe*

- Built and launched a booking and payments platform supporting **108 registered accounts** and **133 scheduled lessons**, processing nearly **CAD $5K within about two months** of commercial operation
- Engineered geographic instructor matching, role-based portals, and Stripe webhooks; automated **8 workflows** covering payroll batching, reconciliation, and notifications
- Built campaign attribution, AI-assisted blog/newsletter automation, social publishing integrations, and technical SEO tooling; implemented analytics tracking **4,967 pageviews over 61 days**

Systems Analyst Intern June 2026 – Present

City of Merritt *Merritt, BC*

- Managed **163 assigned IT tickets**, with **135 completed**, spanning Windows troubleshooting, application support, access management and hardware
- Handled **87 onboarding/offboarding-related tickets** covering employee accounts, permissions and equipment provisioning and collection

Projects

AgentMemory GitHub | Live

Open-Source Contributor *TypeScript, Node.js, MCP, SQLite*

- Contributed tested Pi, OpenClaw, and Hermes integration fixes to an AI-agent memory project with 29.7K+ GitHub stars and 2.6K+ forks
- Worked in a TypeScript codebase spanning session recording, workspace context, and semantic retrieval

Generic Alternatives GitHub | Live

Full-Stack Developer *TypeScript, React, Node.js, PostgreSQL*

- Built supplier-matching and procurement workflows backed by structured data processing
- Implemented background automation for ingestion, logistics tasks, and operational analysis

Technical Skills

Languages: TypeScript, JavaScript, Python, Java, SQL, Rust, PowerShell, HTML/CSS
Frameworks: Next.js, React, Node.js, Prisma
Backend & Data: Convex, PostgreSQL, Supabase, REST APIs, Webhooks, OAuth, RBAC, Session Authentication
Cloud & Development: Vercel, Git, GitHub, Docker, Linux, CI/CD, Automated Testing, Scheduled Jobs
Platforms & Integrations: Stripe, Resend, Twilio, Google APIs, Meta Graph API, GA4
IT Systems: Windows Administration, Microsoft 365, SmartDeploy, BitLocker, Device Deployment, Identity & Access Management